Journal of ISAKOS 18 (2026) 101089

Contents lists available at ScienceDirect

Journal of ISAKOS

journal homepage: www.elsevier.com/locate/jisakos

Original Research

A single-tendon semitendinosus autograft results in lower magnetic resonance imaging graft signal compared with a dual tendon autograft: Outcomes from a subgroup of patients undergoing anterior cruciate ligament reconstruction in the DOSTAR randomized controlled trial

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ARTICLE INFO

Keywords:
Anterior cruciate ligament reconstruction
Hamstring autograft
Signal intensity ratio
MRI outcomes
Graft signal

ABSTRACT

Introduction: Magnetic resonance imaging has been used to evaluate graft integration following anterior cruciate ligament reconstruction (ACLR). The DOSTAR study is a prospective randomized controlled trial (RCT) investigating semitendinosus (ST) versus a dual tendon semitendinosus and gracilis (STG) hamstring autograft. The purpose of the present study is to determine whether magnetic resonance imaging (MRI) analysis identifies differences in graft signal between the two autograft options, and whether signal changes between 6- and 12-months.

Methods: Patients from the DOSTAR trial were invited to take part in this MRI study. Patients underwent anterior cruciate ligament (ACL) reconstruction using either semitendinosus or STG autograft. MRI scans were completed at 6- and 12-months post operatively. The MRI scans were assessed to measure signal intensity ratio (SIR) at several regions of interest. Linear mixed models were used to assess the changes over time for MRI variables. Patient reported outcome measures (PROMS) including the International Knee Documentation Committee (IKDC), Lysholm and Cincinnati scores were documented at 6 and 12 months postoperatively. KT-1000 arthrometer measurements were recorded at 6 and 12 months postoperatively.

Results: A total of 49 patients were recruited into this MRI study, with 38 undergoing MRI at both 6 and 12 months. At 12-months, statistically significant differences were seen on T1 sequences in SIR between the two groups; within the tibial tunnel (3.10 vs 2.07, $p < 0.001$), adjacent to the tibial tunnel (2.63 vs 1.88, $p = 0.002$), in the mid portion of the graft (3.16 vs 2.45, $p = 0.009$), and within a freehand region encompassing the intra-articular portion of the graft (3.03 vs 2.46, $p = 0.018$). Differences were not statistically significant adjacent to the femoral tunnel (2.41 vs 2.08, $p > 0.05$) and within the femoral tunnel (3.09 vs 2.89, $p > 0.05$). There were no statistically significant differences in any PROMS or in KT-1000 measurements between the ST and STG groups at 6 or 12 months.

Abbreviations: MRI, Magnetic Resonance Imaging; ACL, Anterior Cruciate Ligament; ACLr, Anterior Cruciate Ligament Reconstruction; DOSTAR, Dual Or Single Tendon Anterior cruciate ligament Reconstruction; SIR, Signal Intensity Ratio; ROI, Region Of Interest; ST, Semitendinosus; STG, Semitendinosus-Gracilis; TT, Tibial Tunnel; ATT, Adjacent to Tibial Tunnel; MID, MID portion of graft; AFT, Adjacent to Femoral Tunnel; FT, Femoral Tunnel; RCT, Randomized Controlled Trial; LET, Lateral Extra-Articular Tenodesis; BMI, Body Mass Index; AM, Anteromedial; ROM, Range of Motion; FOV, Field Of View; PCL, Posterior Cruciate Ligament; PROMs, Patient-Reported Outcome Measures; IKDC, International Knee Documentation Committee; LKS, Lysholm Knee Score; SD, Standard Deviation; ICC, intraclass correlation coefficients; DFPACLR, Donor-site-Related Functional Problems following Anterior Cruciate Ligament Reconstruction; CKRS, Cincinnati Knee Rating System.

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https://doi.org/10.1016/j.jisako.2026.101089
Received 5 January 2026; Received in revised form 18 February 2026; Accepted 19 February 2026
Available online 23 February 2026

Conclusions: Lower SIR was seen in the semitendinosus group. This was observed in T1 and T2 sequences at 6 and 12 months. Graft SIR did not change significantly between 6 and 12 months ($p > 0.05$).
Level of evidence: Level 2, Subgroup analysis of prospective, single-blinded, RCT.

What are the new findings?

- Signal intensity ratio may be used as a surrogate for graft maturation following anterior cruciate ligament reconstruction.
- Graft signal intensity ratio on magnetic resonance imaging at 6 and 12 months was lower when a semitendinosus only hamstring autograft was used compared to a dual hamstring construct.
- There was no effect on patient-reported outcome measures or functional outcomes at 12 month follow-up.
- This contributes to the body of evidence supporting a single tendon semitendinosus autograft as an option in anterior cruciate ligament reconstruction.

INTRODUCTION

There has been recent interest in the use of magnetic resonance imaging (MRI) as a proxy for healing and integration of grafts in patients undergoing anterior cruciate ligament reconstruction (ACLR) [1,2]. Reinjury after primary ACLR has been associated with increased signal intensity ratio (SIR) adjacent to and within the femoral tunnel aperture on MRI [3]. The increased signal has been correlated with the histological presence of new hypervascular and hypercellular tissue [4]. The DOSTAR study is a prospective randomized controlled trial (RCT) investigating clinical outcomes in patients undergoing ACLR employing a single tendon semitendinosus (ST) versus a dual tendon semitendinosus and gracilis (STG) hamstring autograft, harvested from the ipsilateral limb [5]. The purpose of the present study was to evaluate a subgroup of patients enrolled in the DOSTAR RCT to determine whether MRI analysis at 6 and 12 months demonstrated statistically significant differences in graft SIR between the groups. It was hypothesized that: (1) MRI graft signal in regions of interest (ROIs) on both T1 and T2 sequences would not be significantly different between 6 and 12 month imaging; (2) there would be no statistically significant differences in MRI graft signal between the ST and STG groups on both T1 and T2 sequences. Patient-reported outcome measures and graft laxity measurements were included at 6 and 12 month time points to demonstrate that the sample of patients in each group were clinically comparable and to ensure there was not a large number of clinical outliers in either group. It was hypothesized that there would be no statistically significant differences in patient reported outcome measures (PROMS) outcomes or graft laxity measurements between the groups.

METHODS

Participants

A cohort of patients was identified at the time of their recruitment to the DOSTAR RCT [5]. The trial was prospectively registered on the Australian New Zealand Clinical Trials Registry (ACTRN12620000927921, https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=380071, registered 17/09/2020). Ethics approval was obtained from the University of Western Australia Human Research Ethics Committee (RA/4/20/5941). During the recruitment of patients to the main study, each patient was asked if they consented to involvement in the present imaging study, which involved additionally undergoing MRI at both 6 and 12 months postoperatively. Recruitment to the present MRI-based study aimed to recruit 20 patients undergoing both 6 and 12 months MRI scans in each group. A priori sample size calculation was not performed; availability of funding dictated the number of patients included in this subgroup of the main study. To allow for attrition, recruitment continued until 49 patients had consented to their involvement (October 2021–May 2023). The detailed method of the RCT has been previously published [5]. In summary, all patients underwent ACLR employing a hamstring tendon autograft by the lead author in a single institution (PDA). Patients were randomized to two groups using a randomization schedule provided by a separate member of the research team who was not involved with patient assessment or management. The surgeon was informed of the treatment group allocation just prior to the surgery. Patients were eligible to participate if they had an injury within the last 12 months, had an ACL deficient knee requiring reconstructive surgery using a hamstring autograft and were skeletally mature between 16 and 50 years of age (Table 1). Patients requiring additional procedures such as meniscal repair were included as were those deemed candidates for lateral extra-articular tenodesis (LET). Patients were not eligible to participate if they were unable to provide informed consent, were unable to follow the designated rehabilitation protocol, had a body mass index (BMI) $\geq$ 40, had significant concomitant chondral pathology likely to require intervention, had a multiligament knee injury requiring a second ligament repair or reconstruction, were undergoing a revision ACLR or were unable to undergo MRI scanning. Patients with established osteoarthritis (Outerbridge grade 3 or above) were not included.

Table 1
Participant demographics.

Variable	STG (n = 20)	ST (n = 18)
Age, years	27 (20.5–31.5)	30.5 (24–38)
Sex, n (%)		
Male	8 (40.0%)	13 (72.2%)
Female	12 (60.0%)	5 (27.8%)
Height, cm	173.1 ± 9.9	176.8 ± 7.9
Weight, kg	74.7 ± 14.7	80.4 ± 10.2
Body mass index	24.0 (22.5–26.3)	25.4 (24–27.5)
Operated limb, n (%)		
Right	11 (55.0%)	6 (33.0%)
Left	9 (45.0%)	12 (67.0%)
Time to surgery, weeks	7.5 (3.5–11)	8 (6–14)
Graft diameter, mm	8.5 (8.25–9)	8.5 (8–9)
Tenodesis, n (%)	2 (10.0%)	0 (0%)
Meniscus repair, n (%)	9 (45.0%)	8 (44.4%)
Injury mechanism, n (%)		
Noncontact	18 (90.0%)	16 (88.9%)
Contact	2 (10.0%)	2 (11.1%)

STG = semitendinosus and gracilis dual tendon group; ST = semitendinosus only group; Data reported as mean ± standard deviation, median (interquartile range), or frequencies (n).

Surgical technique

All surgical procedures were standardized except for the graft harvest, graft preparation, and tibial fixation. All patients underwent general anesthesia in a supine position. Patients underwent a standardized regional anesthetic regime consisting of an adductor canal block using local infiltration of 20 ml of 0.2% ropivacaine, placed under ultrasound guidance once the patient was under general anesthetic. A further 50 ml of 0.2% ropivacaine was infiltrated along the graft harvest site by the surgeon, and 20 ml of 0.2% ropivacaine was infiltrated around the wounds at the end of the procedure.

For both groups, there existed an intraoperative exclusion opportunity to ensure the assigned technique allowed a graft diameter of $\geq$ 8 mm. In cases where 8 mm was not achieved with the assigned technique, the patient was excluded from the study and an alternative technique was used to ensure a graft diameter $\geq$ 8 mm was achieved. In the ST group this involved the addition of gracilis and in the STG group this involved conversion to a dual suspensory short graft construct. If randomized to the ST graft construct, the graft was harvested via a transverse incision over the pes anserinus. The semitendinosus tendon was harvested using a closed tendon harvester, quadrupled, and prepared. The graft was fixed with a fixed loop endobutton (Smith and Nephew) at the femoral side, and an adjustable-loop ultrabutton (Smith and Nephew) at the tibial end to cater for the shorter graft. If randomized to the STG graft construct, the grafts were harvested via a single transverse incision over the pes anserinus. Both STG tendons were harvested using a closed tendon harvester and then folded 2–3 times and prepared. The graft was again fixed with a closed-loop endobutton at the femoral end, and a Bio RCI interference screw (Smith and Nephew) 0.5–1 mm greater than graft diameter at the tibial end. The screw was placed anterior to the graft and inserted until the screw head was flush with the tibial cortex.

All grafts were placed on a proprietary tensioner (X-Wing, Smith, and Nephew), set at 50 N tension prior to implantation. Once constructed, grafts were wrapped in a gauze swab soaked in 500 mg of vancomycin, diluted to 20 mL. Knee arthroscopy was performed, with treatment of any chondral or meniscal injury as dictated by the intraoperative findings. The native ACL was resected. Femoral tunnels were prepared using inside-out drilling through the anteromedial portal aiming for an anteromedial (AM) bundle position. Tibial tunnels were prepared using outside-in drilling from the tibial cortex.

Postoperative management

Rehabilitation was surgeon-directed in both groups during the first 6-weeks. Initially, all patients without meniscus repairs were permitted to weight bear as tolerated using crutches. Meniscus repairs were protected with a hinged knee brace and partial or nonweight bearing depending on the repair configuration, with specific restrictions made in the first 6 weeks. Meniscus repairs with a radial element remained nonweight bearing for 6 weeks. Other configuration without disruption of circumferential collagen fibers were permitted to partial weight bear with crutches. Immediate exercises focused on regaining full knee range of motion (ROM), patella mobility, and pain/swelling reduction. Rehabilitation was not standardized from 6 weeks postsurgery and was undertaken under the direction of the patient's own therapist.

Clinical evaluation

Baseline patient characteristics were collected including patient demographics (age, sex, body weight, and BMI) and injury history (time from injury to surgery, leg dominance, injury mechanism). The PROMs were undertaken pre surgery and at 6 and 12 months postsurgery including the International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form, the Lysholm Knee Score (LKS), and the Cincinnati Knee Rating System (CKRS). The KT-1000 knee arthrometer (MEDmetric Corp.) was used at 6 and 12 months post-surgery to measure anterior tibial translation (mm) during a maximal manual test. Investigators involved in clinical evaluation were blinded to treatment group.

MRI evaluation

An examination using a 1.5 T Siemens MAGNETOM MRI scanner was performed at 6- and 12-months postoperatively, using the same machine and protocol. The machine used a 15-channel knee coil. The protocol included high-resolution T1-weighted and T2-weighted images with fat saturation (T2FS) in axial, sagittal, and coronal planes. Sagittal and coronal T1-weighted and T2FS imaging parameters included a 140 mm field of view (FOV) with 17 slices at 2.5 mm slice thickness (voxel size of $0.4 \times 0.4 \times 2.5$ mm^3), with T1-weighted imaging TR = 400 ms and TE = 11 ms and T2-FS imaging TR = 3140 ms and TE = 60 ms. Axial imaging parameters included 140 mm FOV with 17 slices at 2.5 mm slice thickness (voxel size of $0.4 \times 0.4 \times 3.0$ mm^3), with TR = 3020 ms and TE = 58 ms.

The SIR (mean graft signal divided by mean posterior cruciate ligament [PCL] signal) was measured at several ROIs using methodology consistent with the previous literature [1,6–8]. Measurements were taken using both T1 and T2 weighted images, viewed using InteleViewer version 5.2.1.P237 (Intelerad Medical Systems Incorporated, Canada). An 8 mm diameter circular ROI was used to identify mean graft signal at various points using sagittal MRI slices (Fig. 1). Positions used were consistent on 6 and 12 month MRIs. The first ROI was placed within the femoral tunnel (FT) (Fig. 1a), followed by the intra-articular graft adjacent to the femoral tunnel (AFT) (Fig. 1b), at the middle of the intra-articular portion of the graft (MID) (Fig. 1c), in the intra-articular graft adjacent to the tibial tunnel (ATT) (Fig. 1d), and within the tibial tunnel (TT) (Fig. 1e). A ROI drawn freehand to approximate the visible intra-articular graft (full graft) (Fig. 1c). A ROI covering the width of the PCL was used for the PCL at the level horizontal to the tibial spines (Fig. 1c).

The study cohort that underwent MRI at both 6 and 12 months (n = 38) was analyzed independently by three authors (PD, SO'B, TSWG) for inter-rater reliability. The cohort was then reanalyzed by the first author (PD) for intra-rater reliability after a period of 6 months had passed.

Statistical evaluation

Descriptive statistics are reported as mean $\pm$ standard deviation (SD) for normally distributed continuous variables, as median and interquartile range (IQR) for non-normally distributed continuous variables, and frequencies (percentages) for categorical variables. Normality was assessed using the Shapiro–Wilk test. Inter-rater and intra-rater reliability for MRI-based scores was analyzed using intraclass correlation coefficients (ICCs) with absolute agreement using a 2-way mixed model. The ICC was considered poor for values less than 0.5, moderate for values between 0.5 and 0.75, good for values $\geq$ 0.75 and excellent for values $\geq$ 0.90 [9]. Linear mixed models were used to evaluate changes between groups (ST and STG) from 6- to 12- months post-ACLR, adjusting for sex, age, BMI, meniscus repair status, LET, and graft size as fixed-effect covariates, with each participant included as a random intercept to account for repeated measures within individuals. The interaction between group and time was specifically tested to determine if the pattern of change differed between groups. In the occurrence of significant main or interaction effects, independent *t* tests (for normally distributed data) or Mann Whitney U tests (for non-normally distributed data) were used to investigate differences in the dependent variable between the specific assessment time points. An alpha level of 0.05 was used for all statistical tests. All analyses were performed using IBM SPSS Statistics (version 30.0; IBM Corp, Armonk, NY).

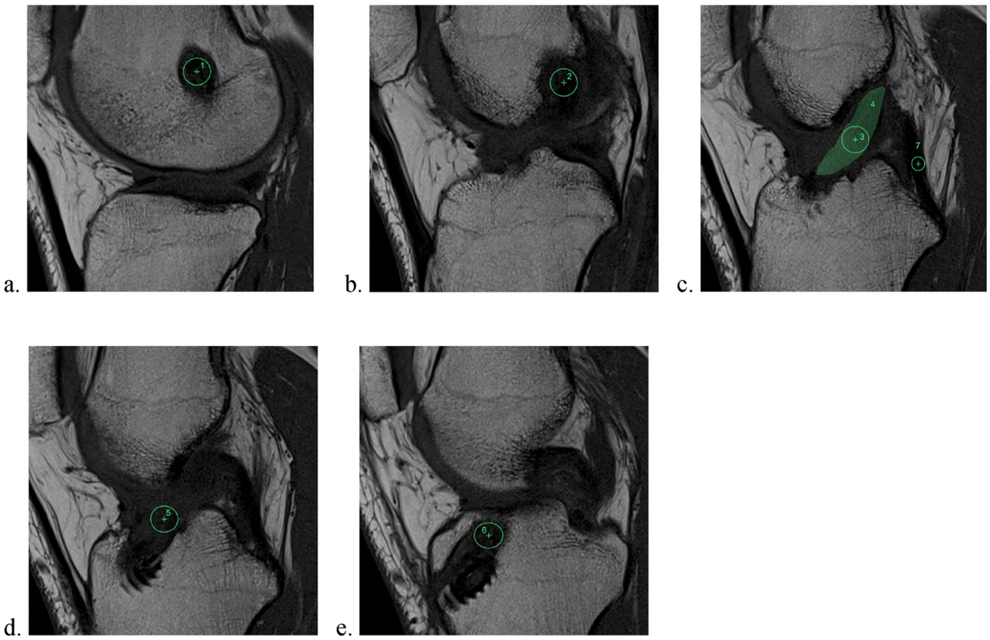

Fig. 1. Location of regions of interest (ROIs). (a) 1 - ROI relating to the graft situated within the femoral tunnel (FT). (b) 2 - ROI relating to the portion of the intra-articular graft adjacent to the femoral tunnel (AFT). (c) 3 - ROI relating to the center point of the intra-articular graft (MID); 4 - ROI relating to a region drawn freehand approximating the intra-articular graft (Full graft); 7 - ROI relating to the posterior cruciate ligament (PCL). (d) 5 - ROI relating to the portion of intra-articular graft adjacent to the tibial tunnel (ATT). (e) 6 - ROI relating to the graft situated within the tibial tunnel (TT). Images were viewed using InteleViewer version 5.2.1.P237 (Intelerad Medical Systems Incorporated, Canada). A circle with a diameter of 8 mm was used to measure ROIs on sagittal images as shown. Additionally, a free-drawn area was used to measure the graft on the slice offering the largest possible area where definite graft tissue could be confirmed. The PCL ROI was smaller to match the width of the visible PCL at the level horizontal to the level of the tibial spine.

RESULTS

Demographics

A total of 38 patients underwent both 6 and 12 month MRI scans and were eligible for inclusion. A flow diagram detailing the recruitment process is provided (Fig. 2). From the 131 patients included in the main study, 49 were consented for involvement in the MRI study (prior to randomization), with 38 patients completing both 6 and 12 month follow-up MRI scans and remaining included in analyses. Details regarding the demographics of the included patients are provided

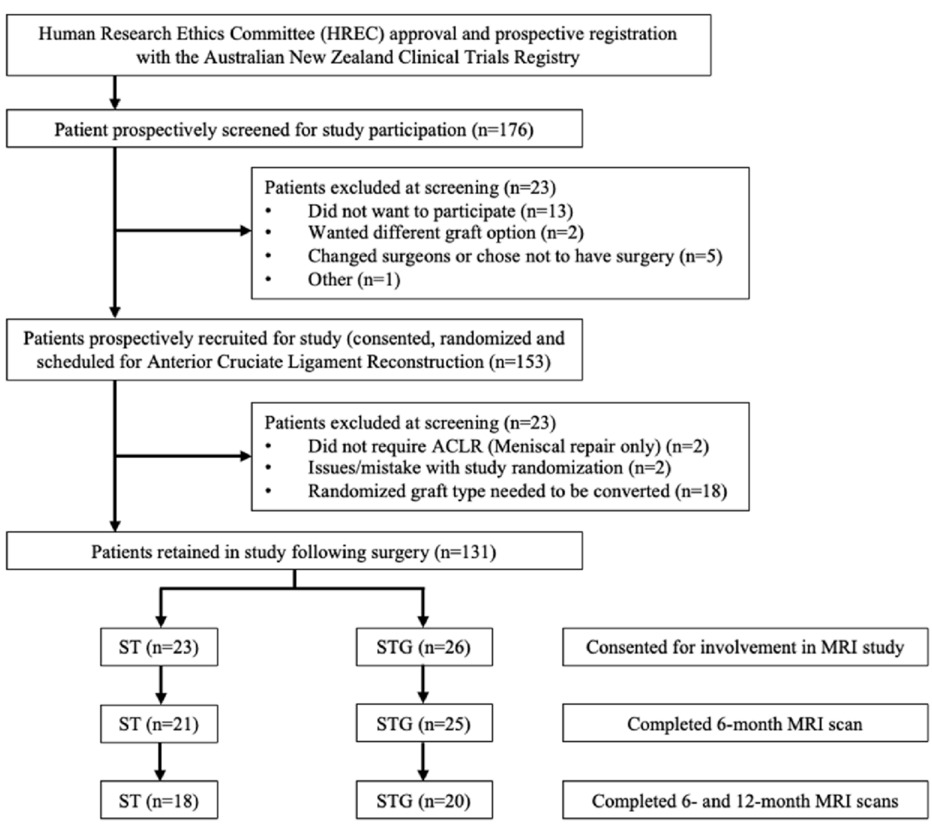

Fig. 2. Flow diagram detailing the recruitment process in patients randomized to the semitendinosus (ST) and semitendinosus/gracilis (STG) autograft groups. ACLR = anterior cruciate ligament reconstruction; MRI = magnetic resonance imaging.

Table 2
Differences in MRI parameters at specific regions of interest between STG and ST graft groups, at 6-month and 12-month assessment using T1 and T2 sequences.

Variable	STG		ST		P value		
	6 months	12 months	6 months	12 months	Time effect	Group effect	Interaction effect
T1 SIR							
TT	3.22 ± 1.34	3.10 ± 0.97	1.92 ± 0.71	2.07 ± 0.66	0.933	**0.002**	0.445
ATT	2.41 ± 0.85	2.63 ± 0.79	1.69 ± 0.61	1.88 ± 0.60	0.080	**0.002**	0.861
MID	3.10 ± 1.28	3.16 ± 0.93	2.18 ± 0.70	2.45 ± 0.60	0.292	**0.014**	0.520
AFT	2.46 ± 0.80	2.41 ± 0.60	2.17 ± 0.83	2.08 ± 0.51	0.563	0.237	0.849
FT	3.21 ± 0.99	3.09 ± 0.79	2.76 ± 0.93	2.89 ± 0.65	0.978	0.524	0.397
Full graft	2.94 ± 0.95	3.03 ± 0.84	2.26 ± 0.70	2.46 ± 0.56	0.304	**0.015**	0.730
T2 SIR							
TT	2.44 ± 0.95	2.11 ± 0.69	1.33 ± 0.53	1.60 ± 0.52	0.761	**<0.001**	**0.011**
ATT	2.17 ± 0.88	2.19 ± 0.80	1.56 ± 0.79	1.62 ± 0.55	0.725	**0.001**	0.864
MID	3.18 ± 1.73	3.21 ± 1.37	1.93 ± 0.97	1.93 ± 0.76	0.943	**<0.001**	0.960
AFT	2.68 ± 1.70	2.32 ± 1.26	1.78 ± 0.78	1.67 ± 0.63	0.296	**0.011**	0.580
FT	3.26 ± 1.28	3.00 ± 1.12	2.52 ± 1.20	2.11 ± 0.73	0.064	**<0.001**	0.655
Full graft	3.09 ± 1.43	3.04 ± 1.13	1.92 ± 0.93	1.97 ± 0.57	0.995	**<0.001**	0.801

MRI = magnetic resonance imaging; STG = semitendinosus and gracilis dual tendon group; ST = semitendinosus only group; FT = femoral tunnel; AFT = adjacent to femoral tunnel; MID = mid portion of the graft; ATT = adjacent to the tibial tunnel; TT = tibial tunnel; Full graft = free drawn area incorporating maximum area of graft. Data reported as mean ± standard deviation and linear mixed model.

(Table 1). There were no statistically significant differences between groups in any demographic measurement.

MRI outcomes

Graft signal measurement demonstrated "excellent" inter-rater reliability (ICC = 0.920 (95% confidence interval [CI]: 0.899–0.936)) and "excellent" intra-rater reliability (ICC = 0.941 (95% CI: 0.923–0.954)). Analysis of T1 weighted MRI sequences at 12 months demonstrated statistically significantly higher SIR in the TT (3.10 vs 2.07, $p < 0.001$), ATT (2.63 vs 1.88, $p = 0.02$), MID (3.16 vs 2.45, $p = 0.009$), and full graft (3.03 vs 2.46, $p = 0.018$) ROIs in the STG group compared with the ST group (Table 2 and Fig. 3).

Analysis of T2 weighted MRI sequences demonstrated statistically significantly higher SIR in all graft locations in the STG group compared with the ST group (group effect ≤0.001) (Table 2). There were no statistically significant changes in graft signal at any ROI between 6- and 12- month MRI scans on either T1 or T2 sequences ($p > 0.05$). Interaction effects were tested and reported in Tables 2 and 3. Only the TT ROI on T2 imaging had a statistically significant interaction effect ($p = 0.011$) (Table 2).

Clinical outcomes

All patients completed clinical follow-up. There were significant improvements in IKDC score, Lysholm score, and Cincinnati score between 6 and 12 month time points (time effect <0.001) but no between group differences ($p > 0.05$) (Table 3). There were no statistically significant differences in KT-1000 laxity between 6 and 12 months or between the STG and ST groups ($p > 0.05$).

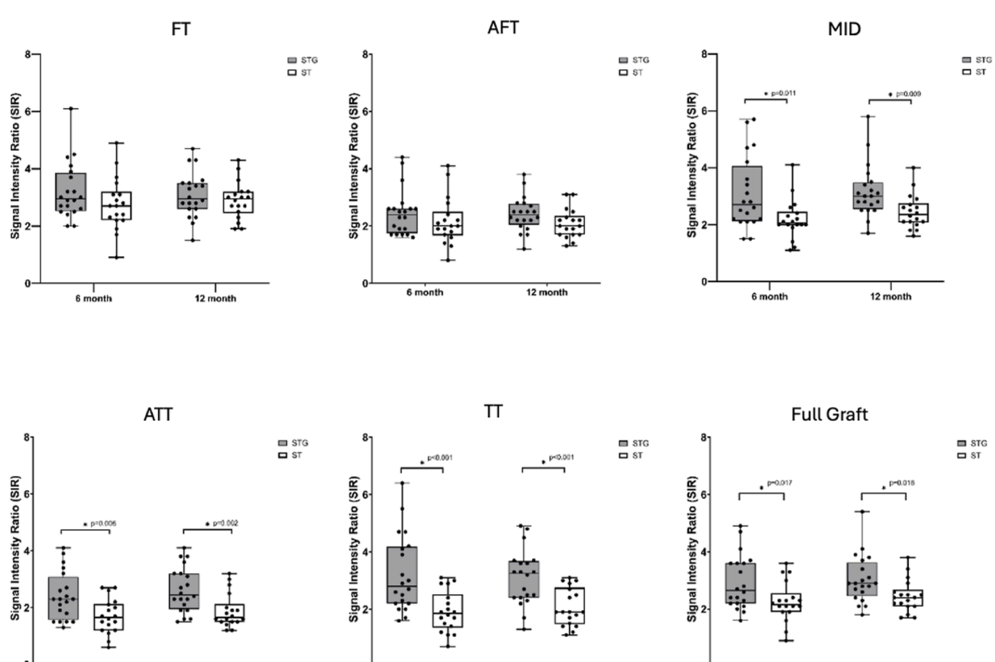

Fig. 3. Box-and-whisker plot comparing signal intensity ratios of different regions of interest for both semitendinosus and gracilis (STG) and semitendinosus only (ST) grafts at 6 and 12 month time points on T1 weighted sequences. FT = femoral tunnel; AFT = adjacent to femoral tunnel; MID = mid portion of the graft; ATT = adjacent to the tibial tunnel; TT = tibial tunnel; Full graft = free drawn area incorporating maximum area of graft.

Table 3
Differences in clinical outcomes between STG and ST graft groups, at 6-month and 12-month assessment post-ACLR.

Variable	STG		ST		P value		
	6 months	12 months	6 months	12 months	Time effect	Group effect	Interaction effect
IKDC	72.7 ± 6.9	86.7 ± 9.0	78.2 ± 9.4	85.2 ± 12.7	**<0.001**	0.825	**0.008**
Lysholm	81.7 ± 8.2	89.3 ± 7.2	85.8 ± 8.1	91.1 ± 11.4	**<0.001**	0.436	0.352
Cincinnati	77.7 ± 7.2	87.9 ± 7.0	80.1 ± 9.6	90.1 ± 10.6	**<0.001**	0.342	0.902
KT-1000 (mm)	1.30 ± 1.26	1.35 ± 0.93	1.11 ± 0.70	1.35 ± 0.84	0.271	0.874	0.461

STG = semitendinosus and gracilis dual tendon group; ST = semitendinosus only group; ACLR = anterior cruciate ligament reconstruction; IKDC = International Knee Documentation Committee Subjective Knee Evaluation Form. Data reported as mean ± standard deviation and linear mixed model.

DISCUSSION

The main findings of this study were, firstly, that there were no significant changes in MRI graft SIR between 6 and 12 months. The literature has been conflicted in demonstrating whether graft signal increases or decreases between 6 and 12 months [8,10]. There is also no consensus as to whether graft signal can predict functional outcome or risk of graft rupture, with a recent study suggesting high MRI signal may be associated with an increased risk of retear of the graft [11,12]. Secondly, graft SIR was significantly lower in the ST group compared with the STG group in all ROIs on T2 sequences, and in the tibial side of the graft on T1 sequences.

This may represent faster “ligamentization” of the ACL grafts when using a single tendon semitendinosus graft as compared with the use of both hamstrings. It may alternatively represent a feature of the fixation methods rather than the graft as lower MRI signal has previously been reported with adjustable suspensory fixation compared with screw fixation [13]. However, the femoral side of the graft was an identical fixation construct in both groups, and differences were still seen on the femoral side. A contributing factor may be the differences found in the microstructure of the ST and gracilis hamstring tendons [14]. The ST has been shown to have a higher density of blood vessels and type 1 collagen compared with gracilis, which has higher density of type III collagen fibrils [14]. The use of 4 strands of ST found in the ST graft may therefore be an advantage, compared with 2–3 strands of both tendons in the STG graft construct, even though the final graft diameter may be similar.

The present study found significant improvements in PROMs between 6 and 12 months as expected during the first year of rehabilitation. No statistically significant differences were found between groups which mirrors the previously published outcomes of the parent study at 6 months [5]. No statistically significant differences were seen in laxity measured by KT-1000 assessment between groups or between 6 and 12 month time points. This also mirrors the outcomes of the parent study at 6 months and confirms that the patients included in the present MRI study had comparable functional outcomes to those in the parent study [5].

The DOSTAR RCT demonstrated significantly higher peak knee flexor strength limb symmetry indices in the ST group ($p < 0.001$), along with less donor site morbidity as measured by the Donor-site-related Functional Problems following Anterior Cruciate Ligament Reconstruction (DFPACLR) ($p < 0.001$) and visual analog scale ($p < 0.001$) [5]. The present study adds to the collection of evidence to support noninferiority of a single ST hamstring autograft compared with traditional methods. Although screw fixation is effective, the advantages of suspensory fixation lie in the versatility of being able to utilize more of the harvested hamstring tendon to create a larger diameter and shorter graft, utilizing the collagen of the ST tendon more effectively [14]. The resultant graft may still be augmented with the addition of a gracilis tendon, although this is generally not required to achieve an acceptable graft diameter of 8 mm [15]. These advantages are useful during training cases, where it is easier for a supervisor to ensure a high-quality autograft. A final benefit is to retain the option of using the gracilis tendon elsewhere. This may be helpful in a multiligament reconstruction where graft options are limited and may prevent the need to use allograft tissue with the associated risks therein.

Some limitations to the present study should be acknowledged. Although the main DOSTAR RCT was double-blinded, this MRI based sub-study could not be blinded to the investigators as the treatment group can be deduced from the appearance of the MRI. Postoperative rehabilitation was not strictly standardized beyond 6-weeks and was undertaken under the direction of the patient's own therapist. The rehabilitation was not recorded and has not been included in the analyses in this study. Although 100% clinical follow-up was achieved with the cohort recruited, three patients failed to attend their initial 6-month MRI and 8 further patients did not attend their 12-month MRI. In total this meant 38 patients attended both MRIs and were included with their paired data. This is a small overall number of patients, but to our knowledge does represent the largest randomized series of its type investigating MRI signal analysis of ST versus STG hamstring autografts.

CONCLUSION

Patients undergoing ACLR using a single tendon ST hamstring autograft with suspensory tibial fixation had reduced MRI graft SIR compared with an STG graft with tibial screw fixation at 6 and 12 months postoperatively. There were no differences in clinical outcomes including PROMS and KT-1000 laxity measurements between treatment groups. This adds to the body of evidence supporting non-inferiority of a single tendon ST hamstring autograft compared with the use of both hamstrings with traditional fixation methods. Long-term clinical follow-up of this cohort is required to ascertain the clinical relevance of these findings.

Patient consent statement

Informed and written consent was obtained from all individual participants included in the study.

Patient and public involvement

Patients and public were not involved in trial design.

Ethics

Ethics approval was obtained by the University of Western Australia Human Research Ethics Committee (RA/4/20/5941). The study was registered with the Australian New Zealand Clinical Trials Registry prior to recruitment (ACTRN12620000927921).

Author contributions

The following authors have conceived and designed the study (JE; PE; PSED; PAD), supervised the conduct of the study (AL; JE; SOB; TSWG; PE; SM; PSED; PAD), analyzed the data (JE; PE; PSED), wrote the initial drafts (JE; PE; PSED), critically revised the manuscript (AL; JE; SOB; TSWG; PE; SM; PSED; PAD) and ensure the accuracy of the data and analysis (AL; JE; SOB; TSWG; PE; SM; PSED; PAD). I confirm that all authors have seen and agree with the contents of the manuscript and agree that work has not been submitted or published elsewhere in whole or part.

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NA.

Data availability statement

Data has not been made publicly available, though data sets generated during the current study can be made available from the corresponding author on reasonable request.

Funding

This study was assisted by a research grant provided by Smith and Nephew Australia (IIS887).

Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Peter D'Alessandro reports financial support was provided by Smith and Nephew Surgical Pty Ltd. Peter D'Alessandro reports a relationship with Smith and Nephew Surgical Pty Ltd that includes: consulting or advisory, funding grants, speaking and lecture fees, and travel reimbursement. Peter Davies reports a relationship with Stryker UK Ltd that includes: speaking and lecture fees. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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